Lecture Notes: Dynamic Contracts and Corruption A Recursive Macroeconomic Approach

Augusto Gonzalez-Bonorino

May 2026

1 Motivation and Positioning

Corruption, sizable informal sectors, and chronically low public investment are jointly determined features of many developing economies. Cross-country data show systematic negative correlations between corruption perceptions and both public capital stocks and formal employment shares (Mauro 1995; Kaufmann et al. 2011; Medina and Schneider 2018). Yet most macroeconomic models address these phenomena in isolation.

This paper develops a unified framework that embeds corruption inside a recursive dynamic contract between a self-interested government and risk-averse citizens who face idiosyncratic productivity shocks and can divert effort into an untaxed informal sector. The government is subject to two-sided *physical* limited commitment: it can expropriate the public capital stock, and citizens can credibly threaten revolution if promised utility falls below a state-dependent outside option. The threshold for revolution depends both on current capital and on an enforcement parameter $\kappa \geq 0$ that captures the state's capacity to monitor, repress, or absorb discontent.

The paper sits at the intersection of four literatures: dynamic optimal taxation under private information, self-enforcing contracts and limited commitment, the political economy of dynamic fiscal policy, and the informal sector. The closest comparator is Acemoglu, Golosov and Tsyvinski (2010), who study dynamic Mirrlees taxation with a self-interested politician disciplined by replacement and prove a separation result. In our environment the politician is disciplined by physical expropriation and revolution, and the per-type Mirrlees wedge is approximately—but not exactly—separable from the corruption multiplier.

2 Environment

Time is discrete and infinite. A continuum of citizens of unit measure and a single self-interested government interact under asymmetric information and two-sided physical limited commitment.

2.1 Preferences and Technology

Citizens have additively separable preferences over formal consumption c and informal time l:

$$U(c,l) = u(c) + v(l) = \frac{c^{1-\sigma}}{1-\sigma} + \psi \frac{l^{1-\gamma}}{1-\gamma},$$

where $\sigma > 0$ is the coefficient of relative risk aversion, $\gamma > 0$ is the inverse elasticity of informal labor supply, and $\psi > 0$ scales the preference for informal activity. Formal labor supply is $e_i = 1 - l_i$.

Each citizen draws an i.i.d. productivity shock $\theta_i \in \{\theta_1 < \cdots < \theta_N\}$ with probabilities $\pi_i > 0$. Output in the formal sector is produced according to the Cobb-Douglas technology

$$y_i = F(K, \theta_i, 1 - l_i) = K^{\alpha}[\theta_i(1 - l_i)]^{1-\alpha},$$

with $\alpha \in (0,1)$. The government observes only formal output y_i; both θ_i and l_i are private information.

Assumption 1 (Single Crossing)**.** *The utility and production functions satisfy*

$$\frac{\partial}{\partial \theta}\left(\frac{v'(l)}{u'(c)F_e(K,\theta,1-l)}\right) < 0.$$

Only local downward incentive constraints bind at the optimum.

2.2 Information and Commitment Frictions

Two-sided physical limited commitment. The government cannot commit not to expropriate. If it deviates it seizes the undepreciated capital stock $(1-\delta)K$ and receives zero continuation value forever:

$$J^{\text{dev}}(K) = (1-\delta)K.$$

Citizens can credibly threaten revolution if promised utility falls below the revolution value. Revolution destroys a fraction $\tau \in (0,1)$ of capital; thereafter citizens operate a constrained-efficient economy whose value is

$$V^{\text{rev}}(K,\kappa) = [1 - h(K,\kappa)] \cdot V^{\text{eff}}(K(1-\tau)),$$

where $V^{\text{eff}}(\cdot)$ is the value function of a benevolent constrained-efficient planner operating the post-revolution economy, and $h(K,\kappa)$ is a non-negative dampening function with $h(K,0) = 0$, $\partial h/\partial \kappa > 0$, and $h(K,\kappa) \in [0,1)$ for all relevant K. Higher κ corresponds to greater state capacity to monitor, repress, or otherwise reduce the appeal of revolution at any given K.

3 The Recursive Problem

The aggregate state is $\Omega = (V, K)$, where V is promised continuation utility to the representative citizen and K is the public capital stock. The government's Bellman equation is

$$J(V, K) = \max_{\{c_i, l_i, W_i\}, S, K'} S + \beta J(\bar{W}, K')$$

subject to

$$\sum_i \pi_i \big(u(c_i) + v(l_i) + \beta W_i\big) = V \tag{PK}$$

$$\sum_i \pi_i c_i + K' + S \leq \sum_i \pi_i F(K, \theta_i, 1 - l_i) + (1 - \delta)K \tag{RC}$$

$$u(c_i) + v(l_i) + \beta W_i \geq u(c_{i-1}) + v(\hat{l}_{i,i-1}) + \beta W_{i-1}, \quad i = 2, \ldots, N \tag{IC}$$

$$u(c_i) + v(l_i) + \beta W_i \geq V^{\text{rev}}(K, \kappa), \quad \forall i \tag{PC}$$

$$S + \beta J(\bar{W}, K') \geq J^{\text{dev}}(K) \tag{GIC}$$

with associated multipliers $\lambda, \rho, \mu_i, \xi_i, \phi$. Here $\bar{W} = \sum_i \pi_i W_i$ is the average promised continuation, and $\hat{l}_{i,i-1}$ is the informal time required for type i to mimic type $i - 1$'s formal output.

Definition 1 (Feasible Allocation)**.** *An allocation $\{c_i, l_i, W_i, K', S\}(V, K)$ is feasible if it satisfies (PK)–(GIC) for all (V, K) in the domain of J.*

Proposition 1 (Existence)**.** *Let the primitives satisfy $\beta(1 - \delta) < 1$, u and v strictly concave, F strictly concave in K and in formal labor, and $V^{\text{rev}}(K, \kappa)$ continuous and increasing in K. There exists a unique bounded continuous function $J(V, K)$ satisfying the Bellman equation, and the optimal policy correspondence is non-empty, compact-valued, and upper hemicontinuous on the domain.*

4 First-Order Conditions

The Lagrangian is standard. The four economically informative first-order conditions are presented below.

4.1 The Corruption Wedge

The first-order condition for stolen rents yields

$$1 - \rho + \phi = 0 \quad \Longrightarrow \quad \rho = 1 + \phi.$$

Whenever the government incentive constraint binds ($\phi > 0$), the shadow price of resources is strictly inflated above unity. This is the corruption wedge: the self-interested government faces an effective marginal cost of public funds equal to $\rho = 1 + \phi$ rather than 1.

4.2 Modified Risk-Sharing

The first-order condition for consumption is

$$\frac{1}{u'(c_i)} = \frac{\lambda}{\rho} + \frac{\xi_i}{\rho\pi_i} + \frac{\mu_i - \mu_{i+1}}{\rho\pi_i}.$$

Because $\rho \geq 1$, every type's consumption is strictly compressed relative to the benevolent-planner benchmark whenever the government's no-deviation constraint binds.

4.3 Inverse Euler Equation

The first-order condition for continuation utility W_i, combined with the consumption FOC and the fact that the (GIC) multiplier ϕ enters the continuation FOC only through $\bar{W}$, yields

$$-J_V(\bar{W}, K') = \frac{1}{u'(c_i)} \quad \text{for all } i = 1, \ldots, N.$$

This inverse Euler relation survives the political-economy frictions exactly as in the benevolent-planner benchmark, provided two scope conditions hold: (i) the government's deviation value J^{dev} depends only on aggregate quantities, and (ii) the participation multipliers ξ_i enter the consumption and continuation-utility FOCs symmetrically. Both conditions are satisfied in our environment.

4.4 The Labor Wedge

Substituting the consumption FOC into the first-order condition for informal time and using the implicit-function relation $\partial \hat{l}_{i+1,i}/\partial l_i = \theta_i/\theta_{i+1}$ yields the central intratemporal optimality condition:

$$\frac{v'(l_i)}{u'(c_i)F_e(K, \theta_i, 1 - l_i)} = \frac{1}{\rho} - \frac{\mu_{i+1}}{\rho\pi_i} \cdot \left(\frac{v'(l_i)}{F_e(K, \theta_i, 1 - l_i)} - \frac{v'(\hat{l}_{i+1,i})}{F_e(K, \theta_{i+1}, 1 - \hat{l}_{i+1,i})} \right). \tag{1}$$

The left-hand side is the citizen's marginal rate of substitution between informal time and formal consumption, scaled by the marginal rate of transformation. At first-best this ratio equals one. The right-hand side decomposes the deviation from first-best into two components: $1/\rho$ captures the corruption wedge (uniform scaling), and the second term captures the standard Mirrleesian informational distortion, scaled by $1/\rho$.

5 Two Political Wedges on Public Capital

Differentiating the maximized Bellman equation with respect to current capital K (envelope theorem) yields

$$J_K(V, K) = \rho \left[\sum_i \pi_i F_K(K, \theta_i, 1 - l_i) + (1 - \delta) \right] - \sum_i \xi_i \frac{\partial V^{\text{rev}}(K, \kappa)}{\partial K} - \phi(1 - \delta). \tag{2}$$

From the first-order condition for next-period capital, $\beta J_K(\bar{W}, K') = 1$, the effective marginal return to public capital that the contract requires for a given K' satisfies

$$\sum_i \pi_i F_K(K^*, \theta_i, 1 - l_i) + (1 - \delta) = \frac{1}{\beta} + \underbrace{\frac{\sum_i \xi_i \partial V^{\text{rev}}/\partial K}{\rho}}_{\text{Revolution Tax (RT)}} + \underbrace{\frac{\phi(1-\delta)}{\rho}}_{\text{Expropriation Tax (ET)}} . \tag{3}$$

Revolution tax. The revolution tax arises because higher current capital raises citizens' outside option $V^{\text{rev}}(K, \kappa)$, tightening their participation constraints. The multiplier $\xi_i \geq 0$ measures the shadow value of relaxing (PC) for type i; the sum $\sum_i \xi_i \partial V^{\text{rev}}/\partial K$ is the aggregate marginal cost of higher capital through this channel. The revolution tax is type-specific in origin but aggregate in the envelope, and is dampened by the enforcement parameter κ.

Expropriation tax. The expropriation tax arises because higher future capital tightens the government's no-deviation constraint $S + \beta J(\bar{W}, K') \geq (1 - \delta)K$ tomorrow. The multiplier ϕ measures the shadow value of relaxing (GIC); $(1 - \delta)$ is the marginal value of K in the deviation payoff. The expropriation tax operates whenever (GIC) binds and is independent of the citizen-side participation structure.

Distinctness. The two wedges are conceptually and mathematically distinct: RT operates through the citizens' participation multipliers and is type-dependent in origin; ET operates through the government's no-deviation multiplier and is purely aggregate. They cannot be collapsed into a single composite wedge by a state-variable transformation.

6 The Labor Wedge and Approximate Separation

Define the labor distortion as the deviation of the marginal-rate-of-substitution-to-marginal-rate-of-transformation ratio from its first-best value of one:

$$D_i \equiv 1 - \frac{v'(l_i)}{u'(c_i) F_e(K, \theta_i, 1 - l_i)}.$$

Substituting the wedge formula (9) and using $\rho = 1 + \phi$ yields the additive decomposition

$$D_i = \underbrace{\frac{\phi}{\rho}}_{\text{aggregate corruption distortion}} + \underbrace{\frac{\mu_{i+1}}{\rho \pi_i} \cdot B_i}_{\text{Mirrlees informational distortion (scaled by } 1/\rho)} , \tag{4}$$

where $B_i \equiv v'(l_i)/F_e(L_i) - v'(\hat{l}_{i+1,i})/F_e(H_{i+1}, \hat{l})$.

Both terms are non-negative under Assumption 1: $\phi/\rho \in [0, 1)$ is monotonically increasing in ϕ, and $B_i \geq 0$ with strict inequality whenever $\theta_{i+1} > \theta_i$. The first term is uniform across types and reflects the inflation of the resource shadow price by the government's no-deviation constraint. The second term is the standard Mirrleesian informational distortion, scaled by $1/\rho$.

Relation to AGT separation. Acemoglu, Golosov and Tsyvinski (2010) prove that in their environment political-economy distortions remain aggregate while per-type Mirrleesian wedges are unaffected. The decomposition (12) is approximately—but not exactly—of this form. The first term ϕ/ρ is uniform across types and aggregate in spirit, consistent with AGT separation.

However, the multiplier $1/\rho$ on the Mirrlees term means that the corruption wedge enters both components, and the equilibrium incentive multiplier μ_{i+1} itself depends on ϕ through general-equilibrium feedback. Strict separation therefore fails.

Lemma 1 (Approximate Separation)**.** *The labor distortion D_i decomposes as in (12). The aggregate corruption distortion ϕ/ρ is the AGT-uniform component; the Mirrlees informational distortion $(\mu_{i+1}/(\rho\pi_i))B_i$ is the type-specific component. Strict separation fails because $\rho = 1+\phi$ enters both terms and μ_{i+1} responds endogenously to ϕ. Numerical experiments across the plausible parameter space show that the aggregate term dominates: D_i is increasing in ϕ for almost all parameter combinations, with the equilibrium response of μ_{i+1} approximately constant or weakly decreasing.*

The numerical analysis underlying Lemma 1 is summarized in Table 1 (full details in Appendix H of the paper). Across 72 parameter combinations, the labor distortion for the low-productivity type is strictly increasing in ρ (equivalently in ϕ). The aggregate component ϕ/ρ accounts for between 63% and 94% of the total distortion at $\rho = 2$.

7 Equilibrium and Comparative Statics

Definition 2 (Stationary Recursive Equilibrium)**.** *A stationary recursive equilibrium is a value function J, policy functions, and a pair (V^*, K^*) such that the policy functions solve the Bellman equation, $\bar{W}(V^*, K^*) = V^*$, $K'(V^*, K^*) = K^*$, and all constraints are satisfied with complementary slackness.*

At any stationary equilibrium, the effective return to capital (11) pins down K^* given the multipliers $\{\xi_i, \phi\}$. Because both the revolution tax and the expropriation tax are non-negative, $K^* \leq K^{FB}$, with strict inequality whenever the corresponding constraints bind.

Comparative statics (conjectures). We do not derive sharp closed-form comparative statics with respect to the deep parameters (ϕ, ψ, κ) for two reasons. First, the model is sufficiently rich that several primitives affect K^* through multiple channels with offsetting signs. Second, the multiplier ϕ is endogenous, so "$\partial K^*/\partial\phi$" is not a well-defined comparative static. We summarize the directions that follow from the envelope condition (11), holding constant the equilibrium response of multipliers, as conjectures to be verified numerically:

1. Tighter (GIC) at the steady state $\Rightarrow$ lower K^*. Holding the participation structure fixed, an increase in the equilibrium ϕ raises the expropriation tax and lowers K^*.

2. Stronger enforcement ($\kappa \uparrow$) $\Rightarrow$ slacker (PC) $\Rightarrow$ higher K^*. Higher κ reduces $V^{\text{rev}}(K, \kappa)$ and weakens the revolution tax.

3. The effect of informal-sector strength ($\psi \uparrow$) on K^* is ambiguous. Larger ψ raises citizens' inside-contract utility, slackening (PC) and reducing the revolution tax through the participation channel. However, larger ψ also raises V^{eff}, potentially raising the outside option V^{rev} and tightening (PC).

8 Conclusion

We have developed a recursive dynamic contracting model in which corruption emerges as the equilibrium outcome of two-sided physical limited commitment between a self-interested government and heterogeneous citizens with private information about their productivity. Three theoretical results stand out: the characterization of two distinct political-economy wedges on the return to public capital (the revolution tax and the expropriation tax), the approximate—but not strict—separation of corruption and Mirrleesian distortions in the per-type labor wedge, and the survival of the inverse Euler equation despite the political-economy frictions.

The framework is parsimonious enough to map to observable cross-country variation in corruption indices, informal-sector shares, and public-investment ratios, and rich enough to nest standard dynamic Mirrlees taxation and limited-commitment risk-sharing as special cases. We view the two-wedge characterization on public capital as the paper's main theoretical contribution. The analysis of approximate separation in the labor wedge is a more modest but novel result: it clarifies that the AGT separation result is robust to several substantial enrichments of the political-economy environment, in the sense that the uniform-scaling component remains the dominant force on labor distortion.